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Cataract detection using Textural Features and Machine Learning Algorithms

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Abstract **According to the World Health Organization report, cataracts are one of the world's leading causes of blindness. Cataract majorly affects the aged population however not restricted to aged people and can be seen among minors too. Among the various types the prominently three types of cataract affect masses in high numbers which are nuclear cataract, cortical cataract and post subcapsular cataract. Conventional methods of cataract diagnoses include slit lamp image test by doctors which does not prove to be effective in classifying cataract in the early stages and can also have inaccuracies in identifying the correct type of cataract. Existing work to automate the process have worked on classification based upon binary detection only or have considered only one type of cataract among the mentioned types for further expanding the system. Our system works on detection of cataract in an attempt to reduce errors of manual detection of cataract in the early ages. Our proposed system has successfully classified images as cataract affected or as a normal eye with an accuracy of 96% using combined feature vectors from SIFT-GLCM algorithm applied to classifier models of SVM, Random forest and Logistic Regression. The effect of using SIFT and GLCM separately have also been studied which leads to comparatively lesser accuracies in the model trained.**

Keywords-Cataract, Image Processing, classification,Texture feature extraction, machine learning.

I. INTRODUCTION

A cataract is a white cloud formation that gradually hardens and forms a yellow sheet on the normally transparent lens of the eye. Cataracts occur when the normally clear lens of the eye becomes cloudy, restricting the amount of light that reaches the retina, leading to a decrease in vision. The World Health Organization conducted a survey on world sight day 2019 under which 36 million people were identified as lined and 217 million had moderate or severe distance vision impairment (MSVI), out of these cases of blindness and MSVI about 65 million people were affected by cataract [1]. Thus, cataract is responsible for or 48% of blindness cases.

Even though cataract develops very slowly it can cause long term problems as it moves from covering a very small portion to spreading over the eye lens causing a vision loss since light can no longer reach the retina due to the cloud that is formed on the lens.

Fig 1 Normal Eye vs Cataract Eye [2]

Among the various signs and symptoms of cataract few of which are prominent include sensitivity to light, double vision, need for brighter light to read and difficulty to see at night [3].

The various causes include diabetes, hypertension, smoking previous eye surgeries, significant alcohol consumption, a family history of cataract and also so the age as one of the important factors. In this project we have taken into consideration the three major types of cataract one of which is nuclear cataract which at the initial stages of development causes nearsightedness however as the time follows the lens becomes densely yellow and also brown in some cases. Moving to the next type which is cortical cataract this begins with the white wedge-shaped opacities on the outer edge of the lens which further moves to the central portion of the lens. Lastly posterior Subcapsular cataract that starts with a small however black colored portion occupying the center of the lens blocking the path of the light to the retina this type of cataracts generally grows faster in comparison to the others [3].

The current system of conventional practices followed to detect cataract or other similar diseases includes a visual acuity test under which the eye sight of the patient is checked with the help of a chart or a viewing device that has letters in decreasing order of its size [3]. This detection technique helps to check the vision accuracy of the patient another method which is majorly used by the doctors and ophthalmologist is slit lamp examination to see the structure of the front of the eye with bright illumination of light on the cornea, iris and lens all under high magnification. Another method is retinal exam in which using a slit lamp or an ophthalmoscope, the doctor examines the lens for signs of cataract.

II. RELATED WORK

In one methodology studied by us proposed a computer-aided system to detect and categorize the cataract using slit-lamp images.
The proposed system performs
1. Lens localization
2. Lens segmentation
3. Feature extraction
4. Categorization.

Lens Localization:
Wide slit illumination is used and images are captured through a digital camera
mounted on a slit-lamp eyepiece. The database of 400 selected images is created and cropped for the further testing process. The cropped and resized image contains the lens structure and part of the iris. It is resized to lower dimensions to obtain a lens circle in a range of radius between 60 to 65 pixels. The hough circle detection algorithm is used to detect the lens center and lens radius.

B. Detection using Support vector machine and statistical parameters
The normalized image contains a random distribution of intensity values. Hence, it can
be considered as a two-dimensional random variables problem.
The random variable is expressed in terms of statistical parameters which were mean, energy, homogeneity, smoothness, contrasts and correlation
A multidimensional support vector machine is used for classification. The SVM model is trained with extracted features of 20 images of each class of nuclear cataracts such as early, intermediate, and mature cataracts. The proposed system stated 90.25 % accuracy in the detection of nuclear cataracts. However, the sensitivity, specificity, and accuracy can be further enhanced by normalizing the segmented pupil or using some other feature extraction technique.[4].

Another paper works upon using the algorithms of Machine Learning and Image Processing together.
The Overall methodology is divided into four steps as follows:
a) Collection of Images
b) Pre-processing of Data
c) Classification by KNN
d) Validation of Developed Application

a) Collection of Images: The data is collected from the google image database by writing a script on JAVA. The Google API using AJAX is being embedded in the application to download the cataract eye images and normal eye images.

b) Pre-processing of Data: The downloaded images are then cropped in order to get only the interesting area of the image. The cropped image is now shown to the user to verify that the image has been cropped appropriately. The image is now fed to the Orange Tool for the process of feature extraction. About 2048 features of the image are extracted with its size and dimensions. Now the dataset is classified into 2 categories Cataract and Normal and both of them are being merged into one .csv file.

Classification by KNN: After the feature extraction process, there is no explicit Training
phase. The KNN model works on Similarity features which are calculated by making use of any distance formula such as Euclidian, Manhattan, etc. The KNN is used in order to decrease the computational work as a mobile application is being used. Based on the similarity, the testing image is classified as either Normal or Cataract.

d) Validation of Developed Application: This is the last stage where the proper working of the mobile application is validated by the users and the Ophthalmologist.
The implementation of the model is done on the Orange tool. Accuracy results of the KNN, SVM, and Naïve Bayes were stated as 83.07%, 72.5%, and 76.67% respectively.[5].

The model for cataract detection, classification, and grading is mainly divided into 3 phases which are described as follows:
a) Pre-processing
b) Feature Extraction
c) Classification
a) Pre-processing: In the Pre-processing stage, the main goal is to remove all the unnecessary information from the image. This is done by resizing the image, cleaning the images. This includes removing glare from the images and also converting the images into grayscale form.
b) Feature Extraction: The main goal of the Feature Extraction process is to filter out the
required features. Now in this situation, the main feature is extracting the image of the eye lens. Therefore, an invisible circle is made and only the part of the image containing the pupil is extracted in this process. After the extraction, it is then given to the wavelet transform with Log Gabor filters to obtain the feature vectors.
c) Classification: This is the last phase of the proposed method wherein the feature vectors which are obtained after performing the Wavelet Transform are sent to both the support machine SVM and ANN. Both of these machines classify the images into three classes: normal, early-stage, and advanced stage. Here the first-class refers to the normal eyes having no sign of cataract, the second class refers to the eyes which are having cataract but it is in the early stages and the last class refers to the eyes in which the cataract has reached the advanced stage.
Classification results show the accuracy of the SVM classifier as 96.8% and the accuracy of the ANN classifier as 92.3% on a dataset of 120 images.[6].

In another research work, the author used digital images of 6 eyes (5 patients) with varying degrees of nuclear cataract, 4 eyes cortical cataracts, and 3 normal eyes. The pictures were converted to greyscale and then to binary formats for finding the relative variations in pixel intensities between healthy and cataractous eyes. The three normal eyes served as a reference.
The lens image is derived from the picture of the patient first. To standardize the comparison and to reduce the calculation load by a third, the RGB image is then converted to a resized greyscale. Compared with the mean grey strength of healthy eyes obtained from 3 natural eyes, the presence of the cataract is then observed. For edge detection and circularity calculation, the image is further converted to black and white. Subsequently, the filtered image is converted to binary format and its circularity ranged from 0.96 to 0.84 for the normal eye and 0.65 to 0.07 for the cataract eye relative to the circularity threshold of the mask. Hence the software was able to detect the presence of cataracts. The automated cataract detection technique displayed accuracy and repeatability, for NC and CC

detection, of
94.96% and 95.14% respectively. [7].

Also, in one of the proposed method digital image processing is used on eye images in this research paper to assess the existence of a cataract along with its severity. A dataset consisting of images of eyes having differing degrees of cataract is applied to two separate image processing algorithms. The first method uses an automatic detection algorithm based on Feature Extraction. This acknowledges the Image pre-processes it, extracts data on the strength of the mean value, and makes decisions on classification into the healthy eye, mild, and severe cataracts.

In an unhealthy eye, the second algorithm is based on measuring the area of the pupil and that of the cataract. The degree of cataract is measured as the relative proportion of cataracts with regard to the pupil region, the affected part. By resizing, converting to grayscale, adding a low pass filter, thresholding, and applying a morphological opening, the image is preprocessed. The transformation of Hough is then used in the picture to define and isolate circles. The transformation of Hough is then used in the picture to define and isolate circles. The picture is reversed using the logical operation of NOT. To detect areas inside or near the pupil, contours are used. If the detected region lies within the detected pupil, a cataract is present. After the presence has been confirmed, areas in both regions are identified and the percentage measured.
Formula used: (cataract area / (pupil + cataract area))* 100 [8].

In one of the research work an algorithm is proposed for cataract screening based on texture features: uniformity, intensity and standard deviation. Their proposed system is as follows [9]:

1. Read true color Image.
2. Convert true color image to grayscale image
3. Enhance contrast grayscale image
4. Apply Gaussian filter to-remove the noise
5. Resize the cropped image size to 256*256
6. Find the histogram of filtered image
7. Calculate the uniformity (U) of image based on histogram values
8. Calculate the mean intensity and standard deviation (sd) of the image
9. Comparison and generation of report

The decision made by the system regarding presence of cataract (Y) and its absence (N) is based upon comparison from set thresholds for mean intensity, uniformity and standard deviation. For a healthy eye, the decision mean intensity threshold value was found to be less than 125.31 and more than 12.93. Uniformity value for a healthy eye was found to be greater than 0.17 and standard deviation was found to be less than 1.59. For cataract suffering patient's eyes, two classes of decisions were made, namely mild and severe. For mild cataracts, mean intensity value was found to be between 125.3 and 134, uniformity value in between 0.09 and 0.11. Standard deviation value was found to lie between 2.64 and 3.18. For severe cataract cases, mean intensity value was found to be more than 134, uniformity value less than 0.09 and standard deviation value greater than 3.18. The proposed method was implemented on MATLAB and gave results with close to 98% accuracy.

III. AUTOMATIC CATARACT DIAGNOSIS MODEL

The following steps are involved in designing and training the model.

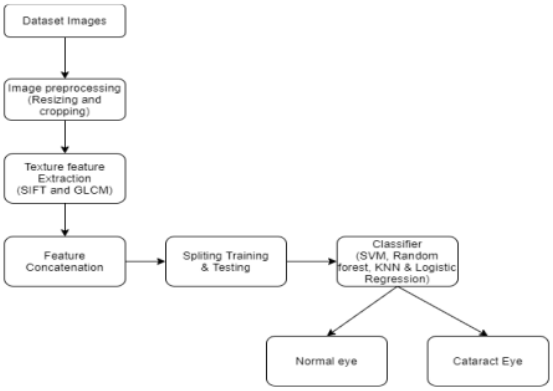

Fig 2 Flowchart of proposed system.

A. *Preprocessing:*

The dataset consists of 97 images that are cropped manually in the initial stage and then they are resized to 128 pixels followed by grayscale conversion of the images which are then used for texture feature extraction.

B. *Feature Extraction using GLCM:*

Gray Level Co-occurrence Matrix (GLCM) is a statistical method of determining texture by calculating the spatial relationship between the pixels. Characterization of texture of the image is done by GLCM by calculating how frequently pairs of pixels with specific values and a specific relationship occur thereby creating GLCM and finally extracting statistical measures from the resultant matrix.

Several statistics can be computed from the gray co matrix. The computed statistical values provide us essential information regarding the texture of the image [10]. The table below lists down some of those statistics.

Table no.1 Statistics Obtained from GLCM

Statistic	Description
Contrast	It measures the local changes in the gray level co-occurrence matrix.
Correlation	It determines the joint probability occurrence of the specified pairs of pixels.
Energy	It determines Angular second moment i.e. it provides the sum of square in GLCM.
Homogeneity	Evaluates how close the distribution of elements are in the GLCM with respect to the GLCM diagonal

C. *Feature Extraction using SIFT:*

Scale Invariant Feature Transform (SIFT) is mainly used to draw out those features from an image which are independent of the scale of image or any kind of rotation. The SIFT algorithm is divided into 4 steps [11]:

1. *Scale-Space Extrema Detection*:

This step focuses on identifying those locations and scales which can be identified from different views of the same object. A "scale space" function is used to achieve those results. For this a Gaussian function is used. The scale space is defined by the function:

$$L(x, y, \sigma) = G(x, y, \sigma) * I(x, y) \quad (1)$$

Here * is the convolution operator, G (x, y, σ) refers to variable-scale Gaussian and I(x, y) is the input image.

Different techniques can then be used to find the locations of stable keypoints in the scale space. One of those techniques is Difference of Gaussian, the scale-space extrema can be found, D (*x*, *y*, σ) by calculating the difference between two images, with one image having scale *k* times the other image. D (*x*, *y*, σ) has the following expression:

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y) \quad (2)$$

For finding the local maxima and minima, each point has to be compared with its 8 neighbouring points located at the same scale, and its 9 neighbours which are present on above and below scale. If the current value has the minimum value of all the neighbouring points then this point is the local minima or if it is maximum then this point is local maxima.

2. *Keypoint Localisation*:

In this step the main aim is to remove more points from the collection of keypoints by finding those having less contrast or are poorly present on an edge. The location of extremum, **z**, is expressed as:

$$Z = \frac{\partial^2 D}{\partial x^2}^{-1} \frac{\partial D}{\partial x} \quad (3)$$

If the resulting value at **z** is less than a threshold value then this point is not considered. This eliminates the extreme values having low contrast. To remove extreme values based on poor localization, the difference of Gaussian function is used. It is generally seen that in these cases there is a huge principle curvature across the edge but a tiny curvature in the perpendicular direction in the difference of Gaussian function. If this difference is less than the ratio of biggest to smallest eigen vector, from the 2x2 Hessian matrix at the location and scale of the keypoint, then the keypoint is excluded.

3. *Orientation Assignment:*

This step focuses on assigning a uniform orientation to the keypoints based on the local properties of the images. The keypoint descriptor, shown below, can then be represented in relation to this orientation, achieving invariance to rotation. The approach taken to find an orientation is:

The keypoints scale is used to select the Gaussian smoothed image L, from above. Orientation histogram is constructed from gradient orientations of sample points. Find the highest peak in the histogram. Using this peak and another local peak within 80% of the height of this peak create a keypoint of that direction. Some points may be having more than one orientation values. To resolve this, a parabola is to be fitted to the 3 histogram values closest to each peak to interpolate the position of peak.

4. *Keypoint Descriptor:*

Keypoint descriptors can also be created using local gradient data. The gradient information is turned around to have the same direction as the orientation of the keypoint and then weighted by a Gaussian with variance of 1.5 * keypoint scale. A set of histograms are created based on the obtained data over a window centred on the keypoint.

Keypoint descriptors generally use a set of 16 histograms, which are aligned in a 4x4 grid, where in each grid consists of 8 orientation bins, one is used for the main compass directions and other is used for each of the midpoints of these directions. This helps in generating a feature vector consisting of 128 elements.

D. *Making a new Feature Vector:*

The features obtained from the above two procedures are then combined together to form the resultant vector which is used in the further steps to train the model. This was done to improve the accuracy of the result by including more features in our vector.

The dataset is then divided into training and testing as 75% and 25% respectively.

E. *Detection and Classification:*

The classifiers used to obtain the results and for the comparative study are as follows

1. *Support vector Machine*:

Support vector machines (SVMs) are supervised machine learning algorithms which are used both for classification and regression problems in which support vectors are the co-ordinates of the individual observation.

SVM mainly focuses on dividing the datasets into classes in order to find a maximum marginal hyperplane (MMH) and it can be done in the following two steps −

- First, SVM will generate hyperplanes iteratively that classify the classes.
- Then, it will choose the hyperplane that separates the classes correctly with minimum outliers.

SVM algorithm is applied with kernel which is used to transform an input data space into the required form. SVM uses kernel which takes a input data space and transforms it into a higher dimensional space.

1.1. Linear Kernel:

Linear Kernel is used when data is linearly separable i.e. it uses straight line to classify the classes. The formula of linear kernel is as below

$$K(x, y) = \text{sum}(x * y) \quad (4)$$

From the above formula, we can see that the product between two vectors say x & y is the sum of the dot product of input values.

1.2. Radial Basis Function (RBF) Kernel:

RBF kernel is mostly used in SVM classification. It maps the input space in indefinite dimensional space. The following formula is used to express it mathematically −

$$K(x, xi) = \exp(-\text{gamma} * \text{sum}(x - xi^2)) \quad (5)$$

Here, gamma ranges from 0 to 1. We can set gamma value in SVM algorithm manually. A gamma value of 0.1 is generally preferred [12].

2. *Logistic Regression*:

Logistic regression is a supervised learning classification algorithm mainly used to predict the probability of a target variable. The nature of the target or dependent variable is binary which means there would be only two possible classes.

In simple words, the dependent variable can have data coded as either 1 (stands for success/yes) or 0 (stands for failure/no).

As a function of x, the logistic regression model predicts P(y=1). The Logistic regression equation can be derived from Linear Regression equation. The steps to get Logistic Regression equations are given below:

Equation of the straight line is written as:

$$y = b0 + b1x1 + b2x2 + b3x3 + \ldots + bnxn \quad (7)$$

In Logistic Regression y can be between 0 and 1 only, so dividing the above equation by (1-y):

$$y/(y-1): 0 \text{ for } y = 0 \text{ and infinity for } y = 1 \quad (8)$$

But we need range between -[infinity] to +[infinity], hence take logarithm of the equation it will become:

$$\text{Log}\,[y/1 - y] = b0 + b1x1 + b2x2 + b3x3 + \ldots.. + bnxn \quad (9)$$

The above equation is the final equation for Logistic Regression [13].

3.*Random Forest*

Random forest is one of the prominent supervised learning algorithms that can be used for both classifications as well as regression. Mainly it is used for classification models. The random forest algorithm produces data decision trees and then gets the predictions from all of these trees and eventually chooses the best solution by voting. It is an ensemble technique that is better than a single decision tree because by averaging the result, it reduces the over-fitting [14].

Step 1 - Initially it starts with a selection of random samples from the dataset.
Step 2 - It constructs a decision tree for each sample and will get the prediction from those decision trees.
Step 3 - Performs voting for every predicted result.
Step 4 - Lastly, selects the most voted prediction as to the final prediction result.

4.*K-Nearest Neighbor*:

K-nearest neighbors (KNN) Algorithm is a type of ML moderated algorithm that can be used for both classifications as well as predictive regression problems. But It is generally preferred for predictive classification problems in the industry[15].

KNN uses 'feature similarity' to predict the values of the latest data points. In simple words, this means that the new data point is assigned a value based on how accurate it is with matches of the points in the training set.

Step 1 – During the initial step of KNN, we must stack the training and the testing data.

Step 2 – Next, we need to choose the nearest data point which is also known as and it can be any integer.

Step 3 – For each point in the test data the algorithm needs to do the following:

3.1 Distance calculation between testing data and each row of training data. Euclidean distance is a commonly used method for calculating distance.

3.2 Sort them in ascending order with respect to their distance.

3.3 Select the first K rows from the sorted array.

3.4 Assigning a class to the test point based on the most frequent class of these rows.

Step 4 – End.

IV. PERFORMANCE OF PROPOSED MODEL

Accuracy is one of the important metrics that help us to determine the performance of our classification or prediction model. It is possible to describe accuracy formally as the number of correct predictions divided by the total number of predictions.
In this research paper we have made an attempt to evaluate the accuracy when different feature extraction algorithms are applied as classification parameters to the machine learning models trained with different algorithms such as SVM-Linear, SVM-Radial basis function, Random forest Logistic regression and KNN. The comparative study of the accuracies can be studied through the tables given below.

Phase 1: SIFT with various classifier algorithms
The table below describes the accuracy of various classification algorithms with the use of SIFT features with SVM algorithm having the highest classification accuracy of 92% along with Logistic Regression. Random forest algorithm achieved an accuracy of 84%. Among all the algorithms used, KNN achieved the lowest accuracy i.e. 68% for cataract classification.

Table No.2 Accuracy rates of various classifiers with SIFT features

Classifier	SIFT
1. SVM-LINEAR	92%
2. SVM-RBF	92%
3. Random Forest	84%
4. Logistic Regression	92%
5. KNN	68%

Phase 2: GLCM with various classifier algorithms

The table below describes the accuracy of various classification algorithms with the use of various GLCM features. Logistic Regression algorithm has achieved the highest classification accuracy of 80%. The SVM-Linear kernel algorithm achieving accuracy of 76% while SVM-RBF achieving the lowest accuracy i.e. 56%. Random forest algorithm achieved an accuracy of 64%. The KNN algorithm has achieved an accuracy of 60% for cataract classification.

Table No.3 Accuracy rates of various classifiers with GLCM features

Classifier	GLCM
1. SVM-LINEAR	76%
2. SVM-RBF	56%

3.	Random Forest	64%
4.	Logistic Regression	80%
5.	KNN	60%

Phase 3: Combination of GLCM and SIFT feature vectors and applied on various classifier algorithms

From the above two feature sets used for classification, it can be observed that the algorithms have achieved better accuracies when used with the SIFT features. Also, the GLCM features have also provided considerable accuracy, which incept the idea of using a new feature set by concatenating the SIFT and GLCM features.
Further in the attempt to improve the accuracies we trained our final model by combining the features from SIFT and G LCM which were 2303 and 5 respectively giving a total of 2308 features in our final vector set.

Table No.4 Accuracy rates of various classifiers with SIFT and GLCM features

Classifier		SIFT + GLCM
1.	SVM-LINEAR	96%
2.	SVM-RBF	96%
3.	Random Forest	96%
4.	Logistic Regression	96%
5.	KNN	56%

The above table shows accuracies obtained by various classification algorithms by using the combined feature set of GLCM and SIFT features. The highest accuracy achieved is 96% by SVM, Random Forest and Logistic Regression and KNN algorithm having lowest accuracy of 56%.

V. DISCUSSION AND CONCLUSION

In this project, various machine learning algorithms have been proposed for the Cataract detection. For this, we have used texture features. We have used a new set of features by concatenating two set of features: SIFT (Scale Invariant Feature Transform) and GLCM (Gray Level Cooccurrence Matrix). Thus, the new set of features have been obtained and used for Cataract prediction. The various algorithms proposed in our project are SVM(Linear), SVM(RBF), KNN, Logistic Regression, Random Forest. Out of which the highest accuracy achieved is 96%. In conclusion, the combined set of features have obtained better results than the individual features.

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